



COURSE:PROBABILITY & STATISTICS
MT-331

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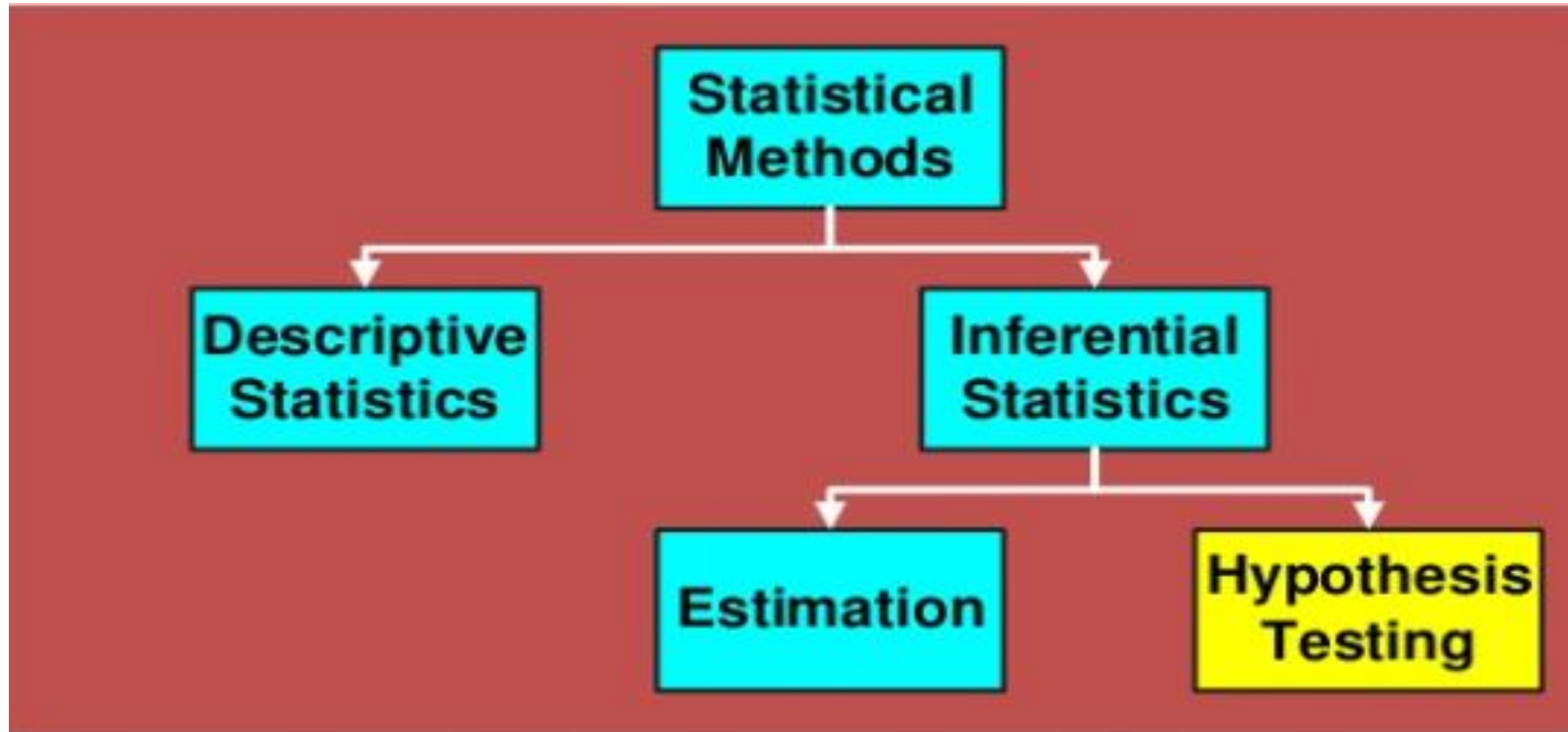
Inferential Statistics

Drawing conclusions and/or making decisions concerning a population based on sample results.

Types of Inferential Statistics

- (a) Statistical Estimation
- (b) Statistical Hypothesis

STATISTICAL INFERENCE



Statistical Estimation

It can be classified in to two categories

- (a) Point Estimation
- (b) Interval Estimation

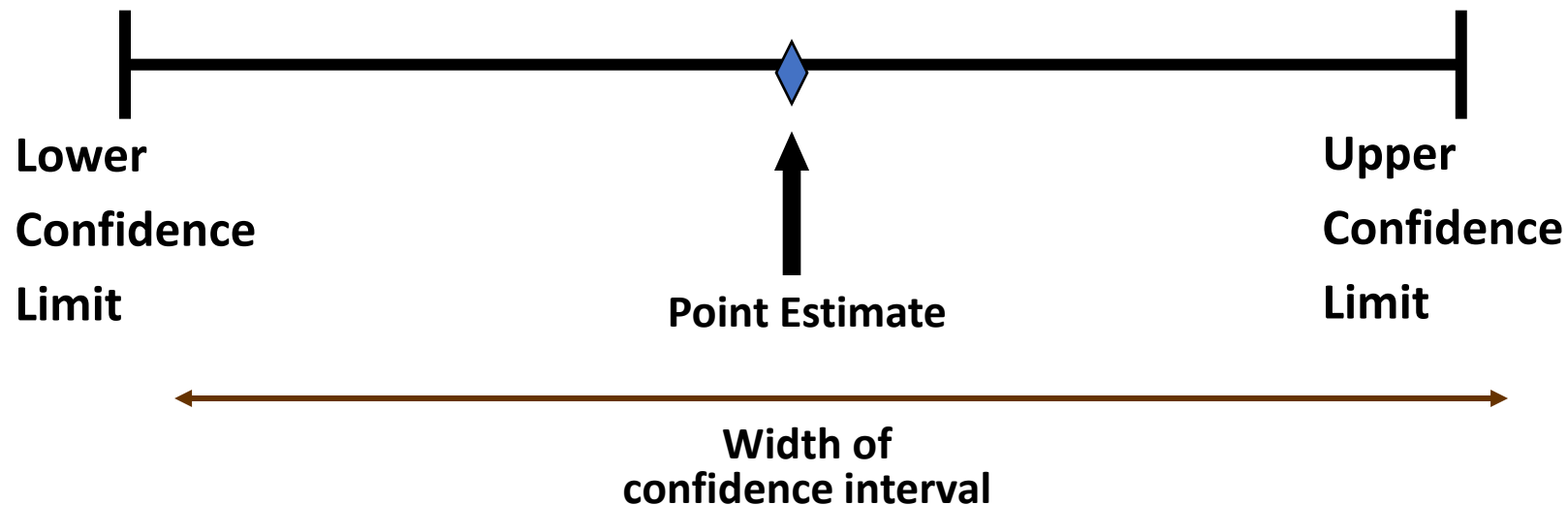
(a) Point Estimation :

When an estimate for the unknown population parameter is expressed by a single value is called point estimate.

(b) Interval Estimation :

Interval estimate means the population parameter given by two number between which the parameter to be lie.

An **interval estimate** provides more information about a population characteristic than does a point estimate. It provides a confidence level for the estimate. Such interval estimates are called **confidence intervals**



INTERVAL ESTIMATION

- CASE 1: σ known , $n \geq 30$

$$\bar{x} - z_{\frac{\alpha}{2}} \left(\frac{\sigma}{\sqrt{n}} \right) < \mu < \bar{x} + z_{\frac{\alpha}{2}} \left(\frac{\sigma}{\sqrt{n}} \right)$$

- CASE 2: σ Unknown , $n \geq 30$

$$\bar{x} - z_{\frac{\alpha}{2}} \left(\frac{s}{\sqrt{n}} \right) < \mu < \bar{x} + z_{\frac{\alpha}{2}} \left(\frac{s}{\sqrt{n}} \right)$$

- CASE 3: σ Unknown $n < 30$ (only sample information given)

$$\bar{x} - t_{\frac{\alpha}{2}} \left(\frac{s}{\sqrt{n}} \right) < \mu < \bar{x} + t_{\frac{\alpha}{2}} \left(\frac{s}{\sqrt{n}} \right)$$

for $z_{\frac{\alpha}{2}}$ & $t_{\frac{\alpha}{2}}$ we use table A3 & A4

Q1-The yield of a chemical process is being studied. From previous experience yield is known to be normally distributed and $\sigma=3$. The past five days of plant operation have resulted in the following percent yields: 91.6, 88.75, 90.8, 89.95, and 91.3. Find a 95% two-sided confidence interval on the true mean yield?

Solution:

$$\bar{X}=90.48,$$

$$n=5 \quad \sigma=3$$

Apply case no 1

$$\bar{x} - z_{\frac{\alpha}{2}}\left(\frac{\sigma}{\sqrt{n}}\right) < \mu < \bar{x} + z_{\frac{\alpha}{2}}\left(\frac{\sigma}{\sqrt{n}}\right)$$

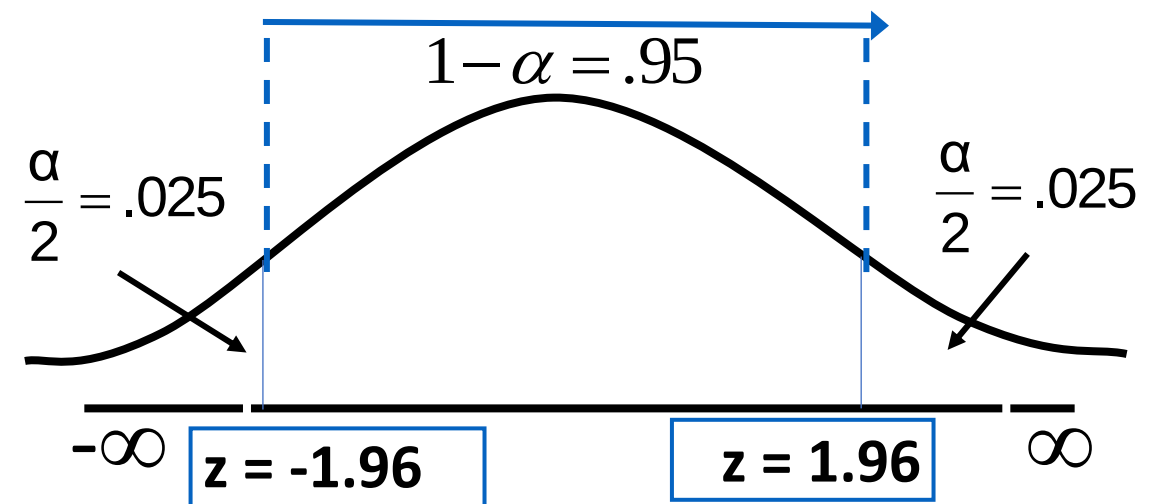
$$Z_{\frac{\alpha}{2}} = Z_{0.05} = Z_{0.025} = 1.96$$

$$90.48 - (1.96)\frac{3}{\sqrt{5}} < \mu < 90.48 + (1.96)\frac{3}{\sqrt{5}}$$

$$87.85 < \mu < 93.10$$

z table A3

z	0	****	0.04	0.05	0.06
*	*	*	*	*	*
*	*	*	*	*	*
*	*	*	*	*	*
-1.9	0.0287	****	0.0262	0.0256	0.025



9.7. A random sample of 100 automobile owners shows that, in the state of Virginia, an automobile is driven on the average 23,500 kilometers per year with a standard deviation of 3900 kilometers. Assume the distribution of measurements to be approximately normal. Construct a 99% confidence interval for the average number of kilometers an automobile is driven annually in Virginia?

Solution :

Sample size = n = 100

Sample mean = $\bar{x}$ = 23500

Standard deviation = S = 3900

99 % confidence interval = ?

$$23500 - (2.57) (3900/ \sqrt{100}) < \mu < 23500 + (2.57) (3900 / \sqrt{100})$$

$$22497.7 < \mu < 24502.3$$

σ unknown , n $\geq$ 30 case 2 apply

$$\bar{x} - z_{\frac{\alpha}{2}}(\frac{S}{\sqrt{n}}) < \mu < \bar{x} + z_{\frac{\alpha}{2}}(\frac{S}{\sqrt{n}})$$

99 % C- I mean α = 0.01

but we not consider the negative sign

$$Z_{\frac{\alpha}{2}} = Z_{\frac{0.01}{2}} = Z_{0.005} = - 2.57$$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049

9.13 A machine is producing metal pieces that are cylindrical in shape. A sample of pieces is taken and the diameters are 1.01, 0.97, 1.03, 1.04, 0.99, 0.98, 0.99, 1.01, and 1.03 centimeters. Find a 99% confidence interval for the mean diameter of pieces from this machine, assuming an approximate normal distribution.

Solution:

Sample size = $n = 9$

Sample mean = $\bar{x} = \frac{\sum_{i=1}^9 x_i}{n} = \frac{9.05}{9} = 1.0055$

Standard deviation = $s = \sqrt{\frac{\sum (X_i - \bar{x})^2}{n-1}} = \sqrt{\frac{\sum (X_i - 1.0055)^2}{9-1}} = 0.0245$

99 % confidence interval = ?

$0.99 = 1 - \alpha = 1 - 0.01$

$\alpha = 0.01$

σ Unknown $n < 30$

$$\bar{x} - t_{\frac{\alpha}{2}} \left(\frac{s}{\sqrt{n}} \right) < \mu < \bar{x} + t_{\frac{\alpha}{2}} \left(\frac{s}{\sqrt{n}} \right)$$

Use table A4

$t_{\frac{\alpha}{2}, n-1} = t_{\frac{0.01}{2}, (9-1)} = t_{(0.005, 8)} = 3.355$

Table A.4
(continued) Critical Values of the t -Distribution

ν	α		
	0.01	0.0075	0.005
1	31.821	42.433	63.656
2	6.965	8.073	9.925
3	4.541	5.047	5.841
4	3.747	4.088	4.604
5	3.365	3.634	4.032
6	3.143	3.372	3.707
7	2.998	3.203	3.499
8	2.896	3.085	3.355

$$\bar{x} - t_{\frac{\alpha}{2}}\left(\frac{s}{\sqrt{n}}\right) < \mu < \bar{x} + t_{\frac{\alpha}{2}}\left(\frac{s}{\sqrt{n}}\right)$$

$$1.0055 - (3.355)\left(\frac{0.0245}{\sqrt{9}}\right) < \mu < 1.0055 + (3.355)\left(\frac{0.0245}{\sqrt{9}}\right)$$

$$0.978 < \mu < 1.0328$$

HYPOTHESIS TESTING

The hypothesis to be tested consists of two complementary statements:

1- The null hypothesis (denoted by H_0) is a statement about the value of a population parameter ; it must contain the condition of equality.

2- The alternative hypothesis (denoted by H_1) is the statement that must be true if the Null hypothesis is false.

e.g :

$H_0: \mu = \text{some value}$	<i>vs</i>	$H_1: \mu \neq \text{Some value}$
$H_0: \mu \leq \text{some value}$	<i>vs</i>	$H_1: \mu > \text{Some value}$
$H_0: \mu \geq \text{some value}$	<i>vs</i>	$H_1: \mu < \text{Some value}$

Level of Significance, α

Defines **rejection region** of the sampling distribution

- Is designated by α , (level of significance)
 - Typical values are .01, .05, or .10
- Provides the **critical value(s)** of the test

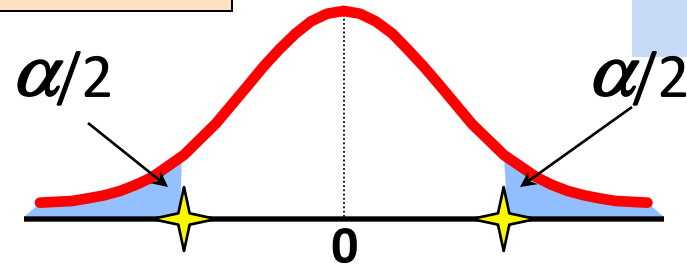
STEP 3: Critical region & Rejection Region

Required Level of significance = α

$$H_0: \mu = 3$$

$$H_1: \mu \neq 3$$

Two-tail test



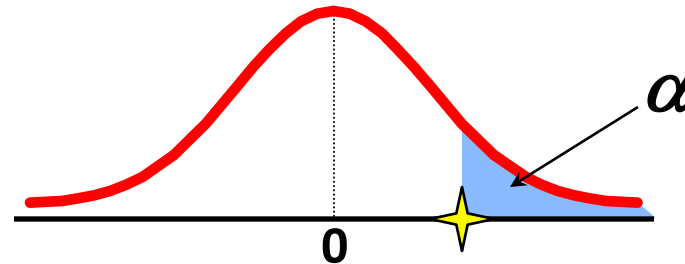
★ Represents critical value

Rejection region is shaded

$$H_0: \mu \leq 3$$

$$H_1: \mu > 3$$

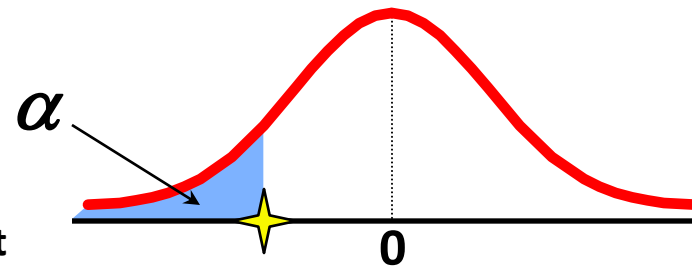
Upper-tail test

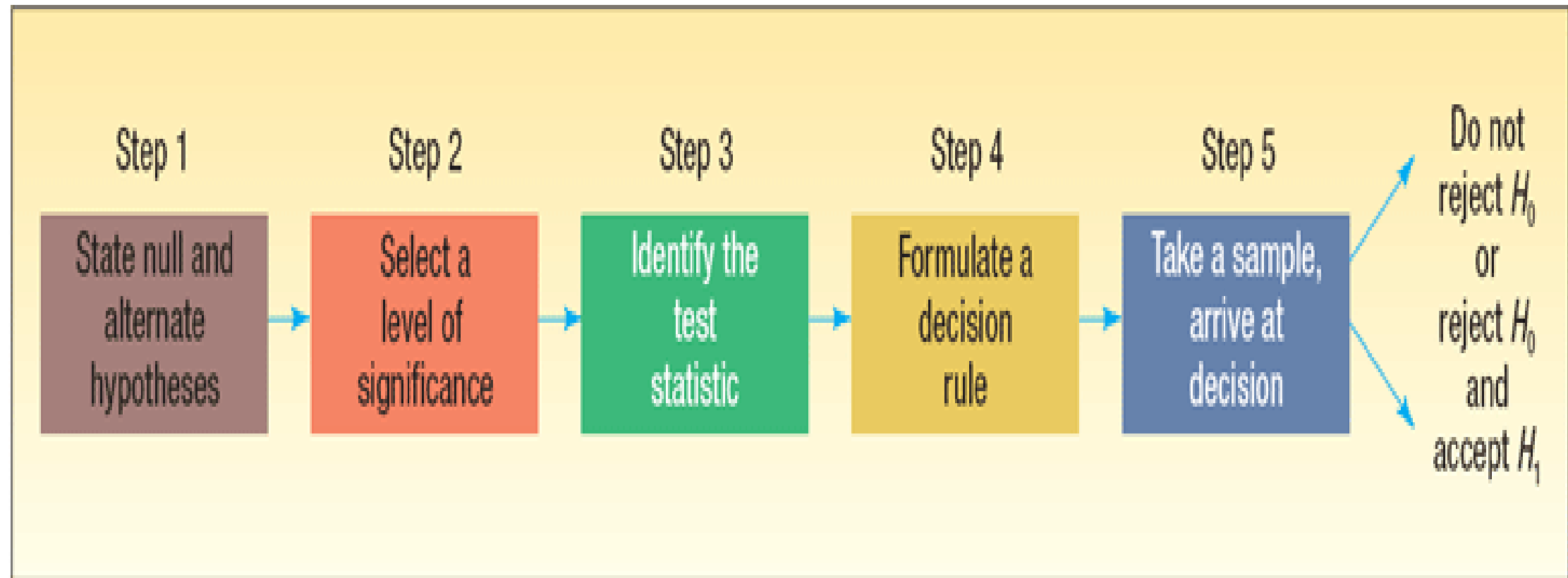


$$H_0: \mu \geq 3$$

$$H_1: \mu < 3$$

Lower-tail test





1- $H_0: \mu = 3$

$H_1: \mu \neq 3 \quad \mu > 3 \quad \mu < 3$

Reject H_0 if $t_{cal} > t_{\frac{\alpha}{2}}$ or $t_{cal} < -t_{\frac{\alpha}{2}}$

Reject H_0 if $z_{cal} > z_{\frac{\alpha}{2}}$ or $z_{cal} < -z_{\frac{\alpha}{2}}$

2- $H_0: \mu \leq 3$

$H_1: \mu > 3$

Reject H_0 if $z_{cal} > z_{\alpha}$ Or Reject H_0 if $t_{cal} > t_{\alpha}$

3- $H_0: \mu \geq 3$

$H_1: \mu < 3$

Reject H_0 if $z_{cal} < -z_{\alpha}$ or Reject H_0 if $t_{cal} < -t_{\alpha}$

10.19 An electrical firm manufactures light bulbs that have a lifetime that is approximately normally distributed with a mean of 800 hours and a standard deviation of 40 hours. Test the hypothesis that $\mu = 800$ hours against the alternative $\mu \neq 800$ hours if a random sample of 30 bulbs has an average life of 788 hours?

Solution:

Step-1

$H_0: \mu = 800 \quad vs \quad H_1: \mu \neq 800$

Step – 2 (Level Of significance) $\alpha = 0.05$

z	.00	.01	.02	.03	.04	.05	.06	.07
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244

Step – 3 (Test Statistic) Use Z-distribution since σ is known

$$Z = \frac{(\bar{x} - \mu)}{\frac{\sigma}{\sqrt{n}}}$$

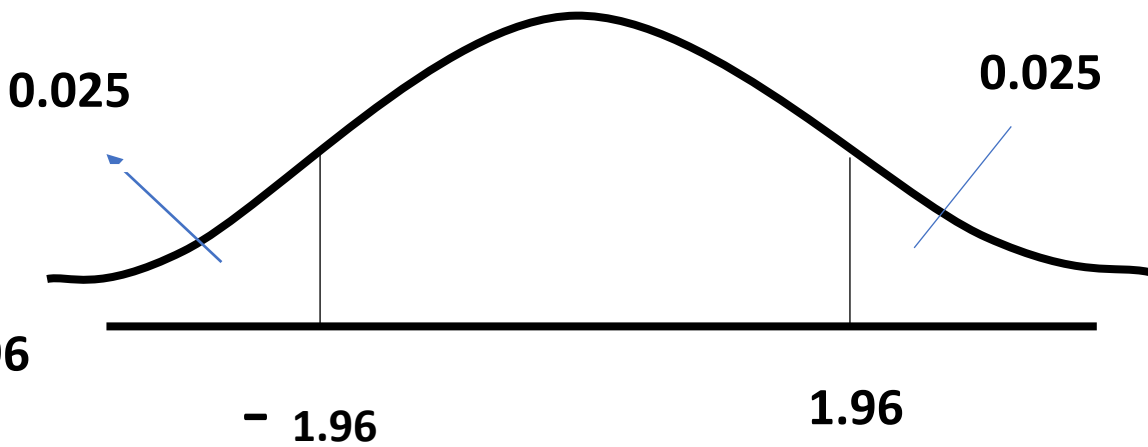
Step – 4 (Critical Region)

Reject H_0 if $z_{cal} > z_{\frac{\alpha}{2}}$

or $z_{cal} < -z_{\frac{\alpha}{2}}$

$$z_{\frac{\alpha}{2}} = Z_{\frac{0.05}{2}} = Z_{0.025} = 1.96$$

$$z_{\frac{\alpha}{2}} = Z_{\frac{0.05}{2}} = Z_{0.025} = -1.96$$



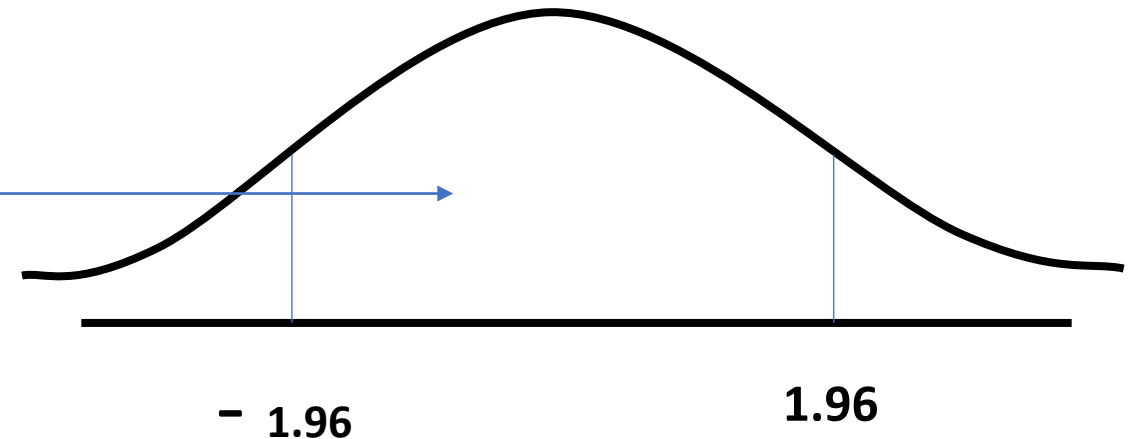
Step – 4 (Critical Region)

Reject H_0 if $z_{cal} > 1.96$ or $z_{cal} < -1.96$

Step – 5 (Calculation)

$$Z = \frac{(\bar{x} - \mu)}{\frac{\sigma}{\sqrt{n}}} \quad \mu = 800, \sigma = 40, n = 30, \bar{x} = 788$$

$$= \frac{(788 - 800)}{\frac{40}{\sqrt{30}}} = -1.64$$



Step – 6 (Decision)

$$1.64 > 1.96 \text{ or } -1.64 < -1.96$$

Accept H_0 and conclude that the average life of light bulbs is equal to 800 hr.

10.20 A random sample of 64 bags of white Cheddar popcorn weighed, on average, 5.23 ounces with a standard deviation of 0.24 ounces. Test the hypothesis that $\mu = 5.5$ ounces against the alternative hypothesis, $\mu < 5.5$ ounces at the 0.05 level of significance?

Solution:

Step-1

$$H_0: \mu = 5.5 \quad vs \quad H_1: \mu < 5.5$$

Step – 2 (Level Of significance) $\alpha = 0.05$

Step – 3 (Test Statistic)

$$Z = \frac{(\bar{x} - \mu)}{\frac{s}{\sqrt{n}}} \quad \mu = 5.5, s = 0.24, n = 64, \bar{x} = 5.23$$

Step – 4 (Critical Region)

$$\text{Reject } H_0 \text{ if } z_{cal} < -z_{\alpha}$$

$$\text{Reject } H_0 \text{ if } z_{cal} < -1.64$$

z	.00	.01	.02	.03	.04	.05
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495

$$z_{\alpha} = Z_{0.05} = -1.64$$

Step – 5 (Calculation)

$$Z = \frac{(\bar{x} - \mu)}{\frac{s}{\sqrt{n}}} \quad \mu = 5.5, S = 0.24, n = 64, \bar{x} = 5.23$$

$$= \frac{(5.23 - 5.5)}{\frac{0.24}{\sqrt{64}}} = -9$$

Step – 6 (Decision)

$$-9 < -1.64$$

Reject H_0 and conclude that the average weight of white cheddar pop corn is less than 5.5 ounces.

10.25 Test the hypothesis that the average content of containers of a particular lubricant is 10 liters if the contents of a random sample of 10 containers are 10.2,9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, and 9.8 liters. Use a 0.01 level of significance and assume that the distribution of contents is normal?

Solution :

Step-1

$H_0: \mu = 10 \quad vs \quad H_1 : \mu \neq 10 \quad (\mu > 10 , \mu < 10)$

Step – 2 (Level Of significance) $\alpha = 0.01$

Step – 3 (Test Statistic)

$$t = \frac{(\bar{x} - \mu)}{\frac{S}{\sqrt{n}}}$$

Step – 4 (Critical Region)

Reject H_0 if $t_{cal} > t_{\frac{\alpha}{2},n-1}$

or $t_{cal} < -t_{\frac{\alpha}{2},n-1}$

$$t_{\frac{\alpha}{2},(n-1)}= t_{\frac{0.01}{2},(10-1)}= t_{0.005,9} = 3.250$$

Table A.4 (continued) Critical Values of the *t*-Distribution

<i>ν</i>	0.0075	0.005
1	42.433	63.656
2	8.073	9.925
3	5.047	5.841
4	4.088	4.604
5	3.634	4.032
6	3.372	3.707
7	3.203	3.499
8	3.085	3.355
9	2.998	3.250

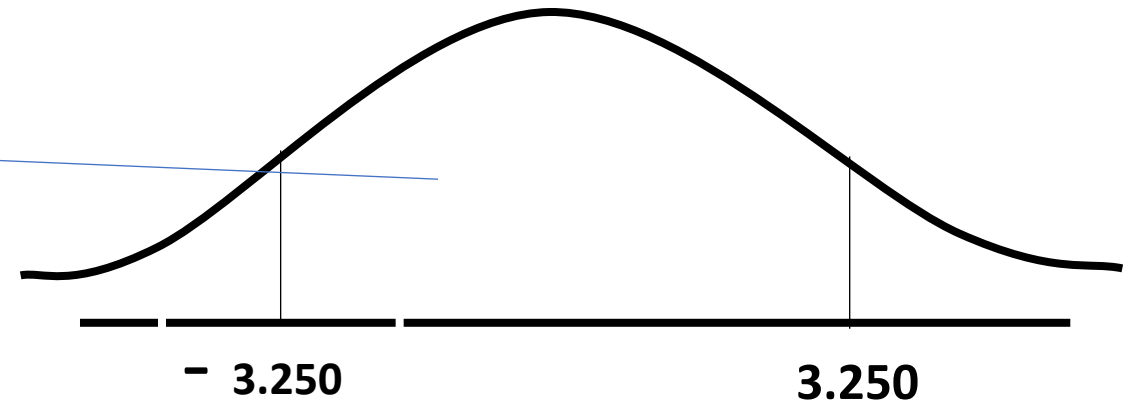
Step – 4 (Critical Region)

Reject H_0 if $t_{cal} > 3.250$ or $t_{cal} < - 3.250$

Step – 5 (Calculation)

$$t = \frac{(\bar{x} - \mu)}{\frac{s}{\sqrt{n}}} \quad \mu = 10, n = 10, \bar{x} = \frac{\sum x_i}{n} = \frac{100.6}{10} = 10.06, s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{(x_i - 10.06)^2}{9}} = 0.2458$$

$$= \frac{(10.06 - 10)}{\frac{0.2458}{\sqrt{10}}} = 0.7719$$



Step – 6 (Decision)

$$0.7719 > 3.250 \quad \text{or} \quad 0.7719 < - 3.250$$

Accept H_0 and conclude that the average content of a particular lubricant is 10 liter.

$$H_0: \mu = \text{value}$$

$$H_1: \mu \neq \text{value}$$

Reject H_0 if:

$$|Z| > Z_{\alpha/2}$$

$$|t| > t_{\alpha/2, n-1}$$

$$H_0: \mu \geq \text{value}$$

$$H_1: \mu < \text{value}$$

Reject H_0 if:

$$Z < -Z_{\alpha}$$

$$t < -t_{\alpha, n-1}$$

$$H_0: \mu \leq \text{value}$$

$$H_1: \mu > \text{value}$$

Reject H_0 if:

$$Z > Z_{\alpha}$$

$$t > t_{\alpha, n-1}$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

10.26 According to a dietary study, a high sodium intake may be related to ulcers, stomach cancer, and migraine headaches. The human requirement for salt is only 220 milligrams per day, which is surpassed in most single servings of ready-to-eat cereals. If a random sample of 20 similar servings of of certain cereal has a mean sodium content of 244 milligrams and a standard deviation of 24.5 milligrams, does this suggest at the 0.05 level of significance that the average sodium content for a single serving of such cereal is greater than 220 milligrams? Assume the distribution of sodium contents to be normal.

Solution:

Step-1

$H_0: \mu \leq 220 \quad vs \quad H_1: \mu > 220$

Step – 2 (Level Of significance) $\alpha = 0.05$

Step – 3 (Test Statistic)

$n = 20, \bar{x} = 244, S = 24.5, \mu = 220$

$$t = \frac{(\bar{x} - \mu)}{\frac{S}{\sqrt{n}}}$$

	α					
v	0.40	0.30	0.20	0.15	0.10	0.05
17	0.257	0.534	0.863	1.069	1.333	1.740
18	0.257	0.534	0.862	1.067	1.330	1.734
19	0.257	0.533	0.861	1.066	1.328	1.729

Step – 4 (Critical Region)

$Reject H_0 \text{ if } t_{cal} > t_{\alpha,n-1} \quad t_{\alpha,n-1} = t_{0.05,20-1} = t_{0.05,19} = 1.729$

$$\text{Reject } H_0 \text{ if } t_{cal} > t_{\alpha, n-1} \quad t_{\alpha, n-1} = t_{0.05, 20-1} = t_{0.05, 19} = 1.729$$

$$\text{Reject } H_0 \text{ if } t_{cal} > 1.729$$

Step – 5 (Calculation)

$$t = \frac{(\bar{x} - \mu)}{\frac{s}{\sqrt{n}}} \quad n = 20, \bar{x} = 244, s = 24.5, \mu = 220$$

$$= \frac{(244 - 220)}{\frac{24.5}{\sqrt{20}}} = 4.380$$

Step – 6 (Decision)

$$4.380 > 1.729$$

Reject H_0 and conclude that the average sodium content for a single serving of such cereal is greater than 220 milligrams

Thank You!

